

Classic Formations: I, Power I, Pro Set, Singleback, and Their Ancestors

The quarterback under center and the backs behind him: eleven formations, what each one was built to do, why most of them faded, and why a few are coming back.

Picture a third-and-1 for the 2024 Baltimore Ravens, late in a close game. The Ravens break the huddle and something about the picture looks old. The quarterback walks right up behind the center and puts his hands under him. Behind him, one directly behind the other, stand a 300-pound fullback and a 247-pound running back, seven yards deep, hands on knees. One tight end, two receivers. The analyst says “they’re in the I,” and the defense, which has spent most of the afternoon with five defensive backs on the field, is already shouting and pointing and walking a safety down toward the line. This is a formation your grandfather would recognize. So why is one of the best offenses in football still using it, and why does the defense react as if it already knows what’s coming? This chapter is about those old shapes: what each was built to do, how they grew out of one another, why most of them faded from the NFL, and why the ones that survived are making a comeback.

You’ll be able to...

- Name the classic formations on sight: the I-formation, offset I, Power I, pro set (split backs), singleback, ace, T-formation, wishbone, flexbone, Wing-T and the goal-line formation.
- Say what each formation was invented to do (give the runner a lead blocker, let him run downhill, stay balanced, or fool the defense with misdirection) and what it gives up.
- Match each formation to the personnel grouping that usually comes with it.
- Explain why two-back, under-center football declined in the NFL, and point to the three places it survives in 2025: 21 personnel, short yardage and play-action.
- Use the backfield as a pre-snap clue: under center or shotgun, one back or two, stacked, split or offset.

Reading a formation from the back to the front

The personnel told you *who* is on the field. The *formation* tells you *where they stand*. Formation Rules and the Language of Alignment gave you the grammar: seven players on the line of scrimmage, the ends of the line eligible to catch passes, the receiver letters (X, Y, Z, H, F), and the quarterback’s three depths, under center (taking the snap directly from the center’s hands), the pistol (about three to four yards back) and the shotgun (about five yards back). This chapter and the next are the vocabulary: the named formations you’ll actually hear on a broadcast.

The classic formations share one thing: **the quarterback is under center**. For most of football history that was simply how the game was played, and the formations differ mainly in what stands *behind* the quarterback. So read them from the back to the front:

1. **Is the quarterback under center?** If he is, you’re in this chapter’s world.
2. **How many backs, and how are they arranged?** One or two or three; stacked one behind another, side by side, or one shifted to the side.
3. **Where are the tight ends and receivers?** A tight end on one side or both, a wing, receivers wide.

Here is the most famous answer to those three questions, labeled part by part.

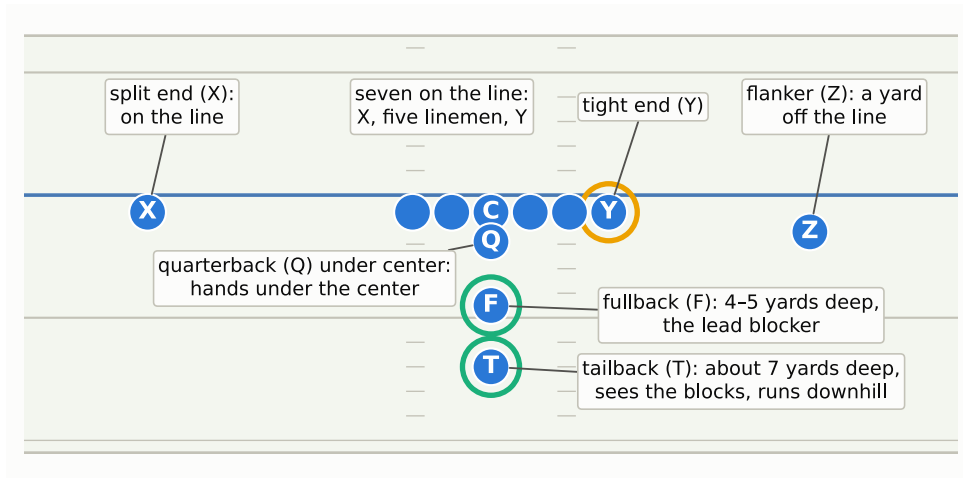


Figure 1: Anatomy of a classic formation: the I-formation (21 personnel). The quarterback (Q) is under center; the fullback (F) stands about 4–5 yards deep, directly behind him; the tailback (T) about 7 yards deep, directly behind the fullback. Seven men are on the line: the split end (X), the five linemen (only the center, C, is lettered) and the tight end (Y). The flanker (Z) stands a yard off the line, so Y, not Z, is the right end of the line and is eligible. Aqua rings mark backs, the gold ring the tight end.

Every formation in this chapter can be described in those same terms, and the names often tell you the shape. The *I* is a letter. The *T* is a letter (the quarterback and three backs form the crossbar and stem). The *wishbone* looks like one. The *pro set* was what the pros used. The naming isn't standardized, and every team has its own playbook words, but the broadcast names are stable enough to learn once.

i Madden vs. real life

In *Madden* you pick a formation from a menu (“I-Form Pro,” “Singleback Ace,” “Strong Close”) and then pick a play from the list that belongs to that formation, as if each formation owned its own plays. Real offenses work the other way around. A team has a few dozen run and pass plays and **runs most of them from many formations**, because a play the defense can predict from the formation is a play the defense can stop. The formation is the first word of the play call (“I Right 26 Power,” as What a Play Really Is spelled out), not the play. The menu names are real, though: *Madden* borrowed them from NFL playbooks.

The ancestor: the T-formation

Almost every formation in this chapter descends from one shape. In the **T-formation** the quarterback stands under center and three backs line up side by side about four or five yards behind him: a fullback in the middle and a halfback on each side. Seen from the press box, the quarterback and the line of backs make a T.

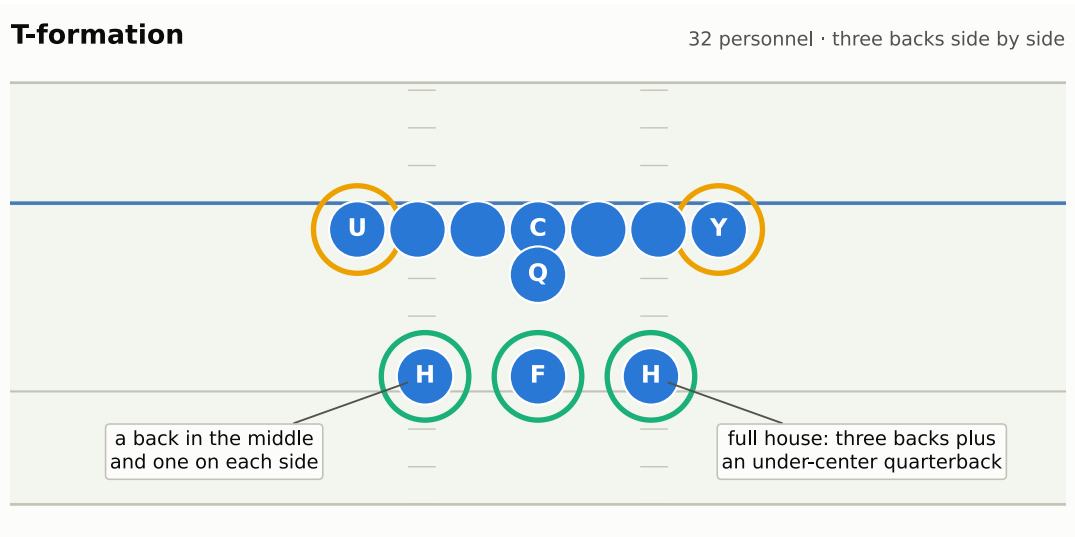


Figure 2: The T-formation (in modern terms, 32 personnel: three backs, two tight ends). Three backs stand side by side about $4\frac{1}{2}$ yards deep, a halfback (H) on each side of the fullback (F), with the quarterback (Q) under center between them and the line. Four players in the backfield behind an under-center quarterback is a full house. Both ends (Y and U) are tight, as in the 1940s; there is no wide receiver at all.

The backs' names are older than the T itself, which is why they're confusing: the fullback is the bigger, blocking-first back, and the halfbacks are the runners. Any backfield with three backs behind an under-center quarterback, four players in all, is called a **full house backfield**. The T is the original full house; you'll meet two more in this chapter (the Power I and the wishbone), and you'll still see one on the goal line most Sundays.

Why it exists: speed to the line

The T's predecessor was the single wing, the dominant formation of the 1910s to the 1930s, in which the center snapped the ball several yards back, through the air, to a tailback who ran, passed or handed off. The single wing is a power formation, but it's slow: a direct snap takes time, the runner starts from a standstill several yards deep, and the defense can see who has the ball.

The T fixed all three problems with one change, **putting the quarterback's hands under the center**. A hand-to-hand snap is instant and almost never fumbled. The quarterback can turn and hand the ball to a back who is already running, so the ball carrier hits the line of scrimmage at full speed one or two seconds after the snap. And because the quarterback can fake a handoff to one back and give to another with his back to the defense, nobody across the line knows for sure who has the ball. With three backs, all a step or two from the line and facing forward, the T threatens the middle and both flanks on every snap, and the defense has to respect all of them.

The modern T arrived with a bang. Coaches had used a tight T for decades, but in 1940 Clark Shaughnessy's version of it, with wider splits, quicker hitting plays and a back sent in motion before the snap, went undefeated at Stanford, and George Halas's Chicago Bears, who had been developing the same ideas with Shaughnessy's help, used it to beat Washington **73-0** in the NFL Championship Game on December 8, 1940.¹ Within a decade almost every team in the NFL had switched; the Pittsburgh Steelers, the last NFL holdout, used the single wing for the last time in 1952 and switched to the T in 1953.² The Single Wing to the Scoring Drought tells that history in full; what matters here is that the T is the trunk of the tree.

What did the T give up? **Width**. With both ends tight and three backs packed behind the quarterback, it had no receivers outside the formation, so the defense could pack its own players near the ball too. Most of what happened next is the story of coaches taking the T and pulling one of its players out wide.

¹"1940 NFL Championship Game," Wikipedia: https://en.wikipedia.org/wiki/1940_NFL_Championship_Game (Bears 73, Washington 0, Dec. 8, 1940); "1940 Stanford Indians football team," Wikipedia: https://en.wikipedia.org/wiki/1940_Stanford_Indians_football_team (9-0 in the regular season, then a 21-13 Rose Bowl win over Nebraska on Jan. 1, 1941, a year after going 1-7-1); "1941 Rose Bowl," Wikipedia: https://en.wikipedia.org/wiki/1941_Rose_Bowl (Shaughnessy had worked informally with George Halas on the Bears' T and was still a Bears consultant when he helped prepare them for the 1940 title game).

²"1952 Pittsburgh Steelers season," Wikipedia: https://en.wikipedia.org/wiki/1952_Pittsburgh_Steelers_season ("the last year the Steelers used the single-wing formation on offense, switching to the T formation the following year"; the last NFL team to use it as their primary formation).

The family tree

The diagram below is the map for the rest of the chapter. Each branch is an answer to the question “what should we do with the T’s three backs?” Some coaches stacked them, some split them, some spread them out, some kept all three and taught the quarterback to read the defense instead of blocking it.

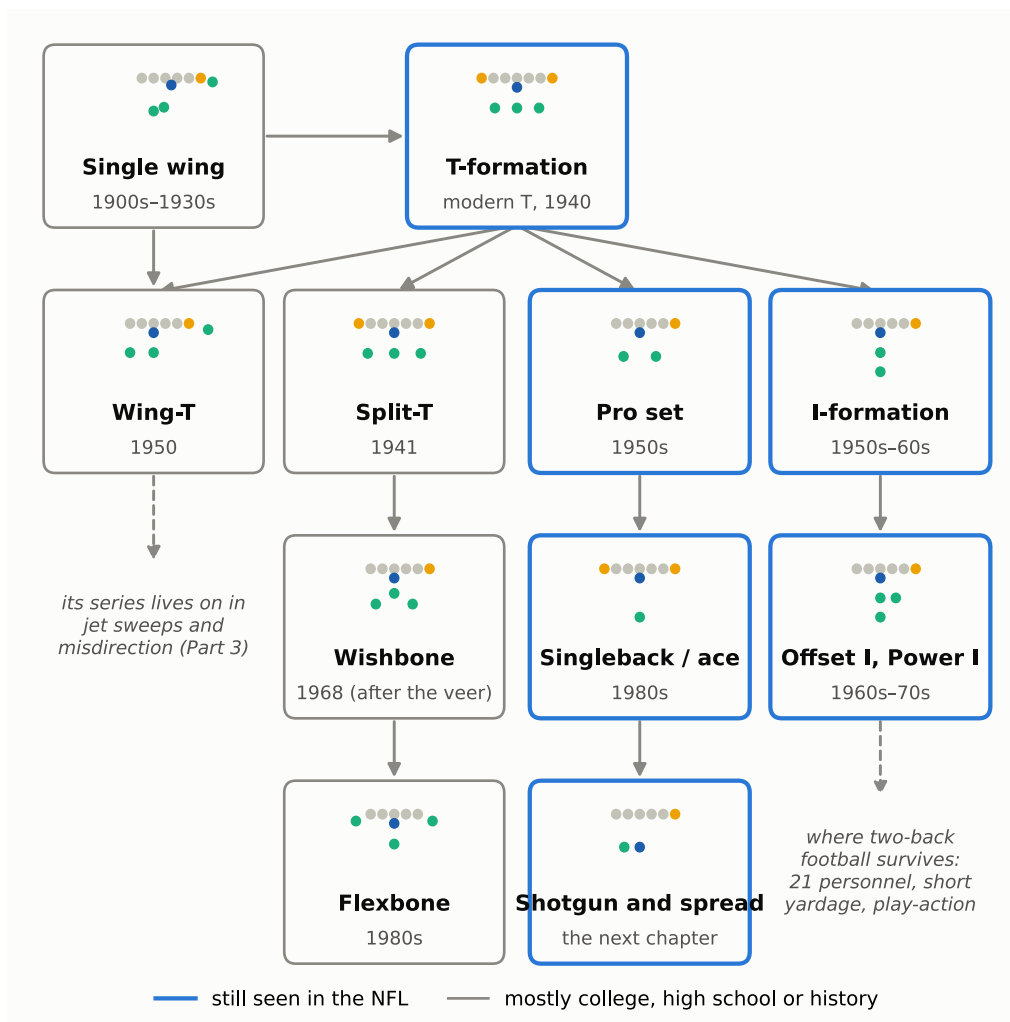


Figure 3: The family tree of the classic formations (dates are when each became prominent, not the first time anyone drew it; several are disputed, see the text). Each icon shows the line (gray), the quarterback (dark blue), the backs (aqua) and any tight end or wing close to the line (gold). Blue boxes are formations you can still see in today’s NFL; gray boxes mostly live on in college, high school and the service academies. The shotgun and spread formations, the trunk’s newest branch, get their own chapter.

You don’t need to memorize the dates. The useful thing is the logic of the branches, because each one is a different answer to the same trade: **every player you keep in the backfield is a blocker or a runner you gain, and a receiver you lose.**

The pro set: one back goes wide

The first big branch came from the passing game. In 1949 Shaughnessy, by then coaching the Los Angeles Rams, took one of his T-formation halfbacks, Elroy “Crazylegs” Hirsch, and lined him up wide as a receiver: a flanker.³ (Others had split a back wide before, so Hirsch’s claim to be the *first* flanker is disputed, but he became the model for the position.) Two backs stayed behind the quarterback, side by side. Add a tight end on one side and a receiver split wide on the other, and you have the formation that ran the NFL for the next quarter century.

³“Elroy Hirsch,” Pro Football Hall of Fame: <https://www.profootballhof.com/players/elroy-hirsch/> (66 catches, 1,495 yards and 17 touchdown catches in 1951); “Elroy Hirsch,” Wikipedia: https://en.wikipedia.org/wiki/Elroy_Hirsch (Shaughnessy, the Rams’ head coach, moved Hirsch to end after he signed in 1949; the Rams often split him wide and later moved him permanently to end, “the first flanker”; the article also notes the dispute, since Don Hutson’s Packers and the 1940s Bears and Washington had split backs wide before).

The **pro set**, also called **split backs**, puts the quarterback under center and the halfback and fullback side by side behind him, usually about five yards deep and roughly behind the guards or tackles. Up front there's a tight end on one side, the flanker (Z) a yard off the line on the tight end's side, and the split end (X) on the line on the other side. That's 21 personnel: two backs, one tight end, two receivers.

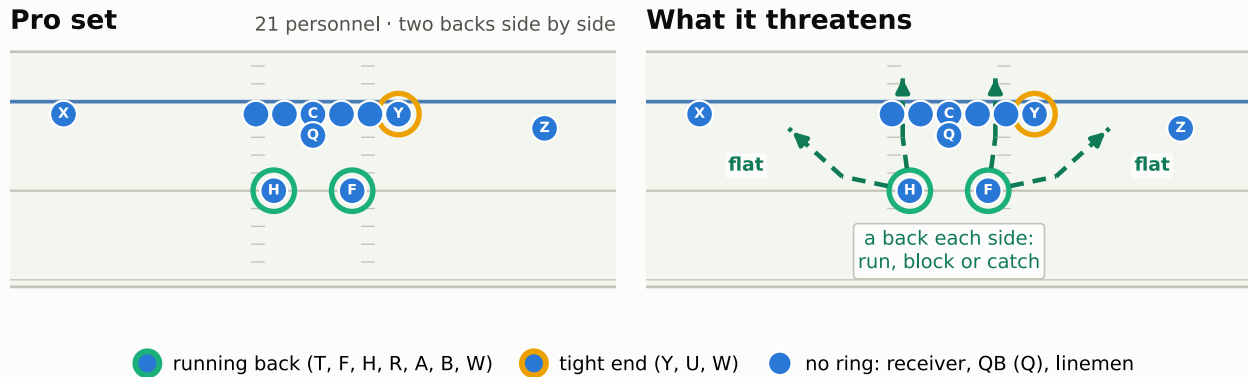


Figure 4: The pro set (split backs, 21 personnel). Left: the plate. The halfback (H) and fullback (F) stand side by side about five yards deep; the tight end (Y) and flanker (Z) are on the right, the split end (X) on the left. Right: what the backfield threatens. Each back can carry the ball on his own side (the dashed green arrows straight ahead) or slip out into a short pass route to his side (the flat), so the defense can't tell the play's direction from the backs' alignment, and both edges of the pass protection have a back beside them to pick up a blitz.

Why it exists: balance

The pro set's virtue is *balance*. With a back on each side of the quarterback, the offense can run either way with a lead blocker, throw to either back in the flat, or keep both in to block a blitz on either side. The defense gets almost no clue from the backfield, because the backfield looks the same whichever way the play goes. Meanwhile the flanker and split end give the passing game two real receivers outside, which the T never had. That combination, a credible run game both ways plus two wide receivers and a tight end, made the pro set the standard NFL formation from the late 1950s into the 1980s. Bill Walsh's early San Francisco offenses, the original West Coast offense, was built on split backs and threw many of its short passes to backs releasing from them.⁴

What does it give up? **A downhill runner.** Neither back is directly behind the quarterback, so neither gets a straight-ahead running start, and a back who takes a handoff from a split alignment is usually running at an angle, sideways first. And the lead blocker, if there is one, has to cross in front of the ball carrier. For power running straight at the defense, the next branch was better.

i Film room: Green Bay Packers, 1959–1967

Vince Lombardi's Packers won five NFL championships in seven seasons (1961, 1962, 1965, 1966 and 1967, the last two followed by wins in the first two Super Bowls), and they did it mostly from the pro set.⁵ Fullback Jim Taylor and halfback Paul Hornung lined up side by side behind Bart Starr, and the signature play was the *power sweep*: both guards (Jerry Kramer and Fuzzy Thurston) pulled toward the sideline, the fullback blocked the defensive end, and the halfback took the handoff wide behind a wall of blockers. Lombardi's name for the formation was "Red": split backs, set wide apart. Lombardi's idea was that the runner shouldn't aim for a fixed hole but follow his blockers and "run to daylight," cutting wherever the defense left an opening; it's the title of his 1963 book.

What to notice: the split backs are what made the sweep hard to diagnose. The same backfield ran the sweep, the fullback's runs inside and the plays that punished defenses for chasing the sweep, and the defense couldn't tell which from the backfield. (The sweep itself, drawn block by block, lives in *The Single Wing to the Scoring Drought*.)

⁴"West Coast offense," Wikipedia: https://en.wikipedia.org/wiki/West_Coast_offense ("originally the offense used two split backs," with the backs' check-release and free-release rules that sent them into short routes).

⁵Pro Football Hall of Fame, "Vince Lombardi": <https://www.profootballhof.com/players/vince-lombardi/> (Packers 1959–1967; NFL championships 1961, 1962, 1965, 1966, 1967; Super Bowls I and II). Vince Lombardi with W. C. Heinz, *Run to Daylight!* (Prentice-Hall, 1963). The formation and blocks: Cliff Christl, "Red Cochran on Lombardi and Hall of Fame backs Hornung, Taylor," Packers.com, March 15, 2018: <https://www.packers.com/news/red-cochran-on-lombardi-and-hall-of-fame-backs-hornung-taylor-20452981> ("Red Right 49"; "the red was the formation with a wide split between the backs"; the fullback on the defensive end; Kramer and Thurston pulling).

The I-formation: stacking the backs

If you want the ball carrier running straight at the defense, put him directly behind the quarterback, and put the blocker in front of him. That is the **I-formation**: quarterback under center, the fullback directly behind him about four to five yards deep, and the tailback directly behind the fullback about seven yards deep. Three players in a straight line make the I. With one tight end and two receivers it's 21 personnel, and "I-formation" and "21 personnel" were close to synonyms for a generation.

The I is usually credited to Tom Nugent at the Virginia Military Institute in 1950 (stacked backfields appear in older playbooks too, so "invented" is disputed). It became famous at the University of Southern California in the 1960s under John McKay, who learned it from his assistant Don Coryell, and whose I-formation tailbacks (Mike Garrett, then O.J. Simpson) won Heisman Trophies and gave USC the nickname "Tailback U."⁶ By the 1970s and '80s it was everywhere, college and pro.

The deep tailback is the whole point. Seven yards is a long way back, and it buys three things.

A running start. The tailback takes the handoff already moving at speed, going straight forward, toward the line: what coaches call running *downhill*. A runner who meets a linebacker at full speed going forward wins more collisions than one who meets him moving sideways.

Time to see. From seven yards the tailback can watch the blocks form in front of him before he commits. If the hole he was promised closes, he can see the one that opened instead.

A lead blocker. The fullback is between the tailback and the line, so on a run he goes first, into the hole, and blocks the first defender who fills it, usually a linebacker. That's a lead block, and the simplest I-formation run of all, the iso (short for "isolation"), is built on it: the line blocks the defensive linemen, the fullback blocks one linebacker, "isolating" him, and the tailback runs off the fullback's block. The iso and its relatives are taught properly in Gap and Power Runs; here it's the cleanest picture of why the I exists.

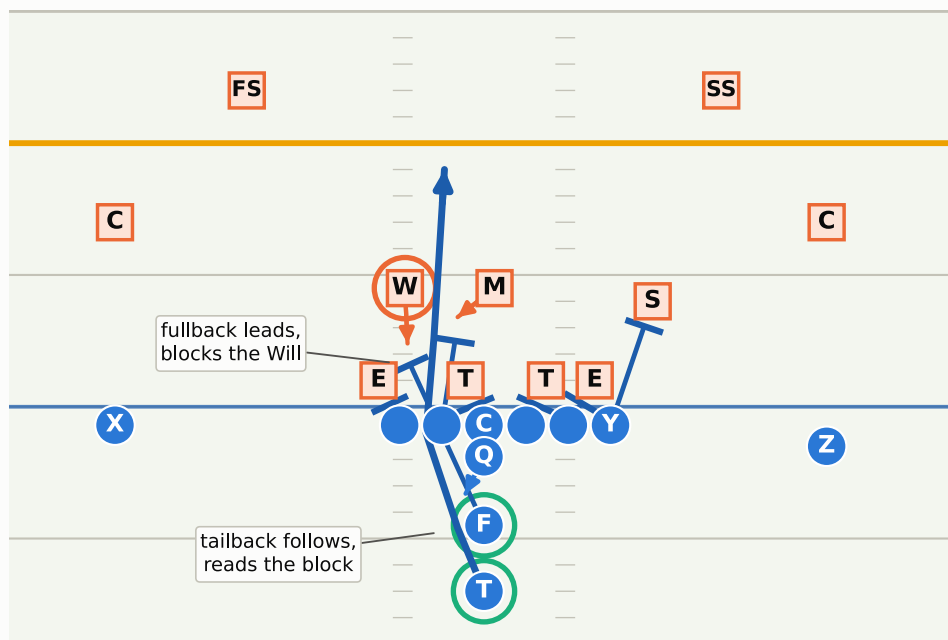


Figure 5: Why the I exists: the iso (isolation) run, drawn against a base 4-3 defense. The five linemen and the tight end each block a defender in front of them (T-bars). The fullback (F, aqua ring) leads straight through the gap between the left guard and left tackle and blocks the Will linebacker (W, orange ring), the one defender nobody else blocked. The tailback (T) takes the handoff about five yards deep, already moving forward, and follows the fullback. Defense: E = end, T = tackle, S/M/W = linebackers, C = cornerbacks, FS/SS = safeties.

⁶"I formation," Wikipedia: https://en.wikipedia.org/wiki/I_formation (Tom Nugent credited with developing it at VMI in 1950; possible earlier sources include Charles Hollister at Northwestern, 1900, and Bob Zuppke, 1914; Don Coryell used the I before joining McKay's USC staff in 1960, and John Madden credited Coryell with bringing it to McKay); "Tom Nugent," Wikipedia: https://en.wikipedia.org/wiki/Tom_Nugent ; Heisman Trophy winners Mike Garrett (1965) and O. J. Simpson (1968): <https://www.heisman.com/heisman-winners/> .

Watch the same play in motion. Notice how far the tailback has to travel before he gets the ball, and how that distance is what lets him arrive at the line of scrimmage at full speed, a split second after the fullback.

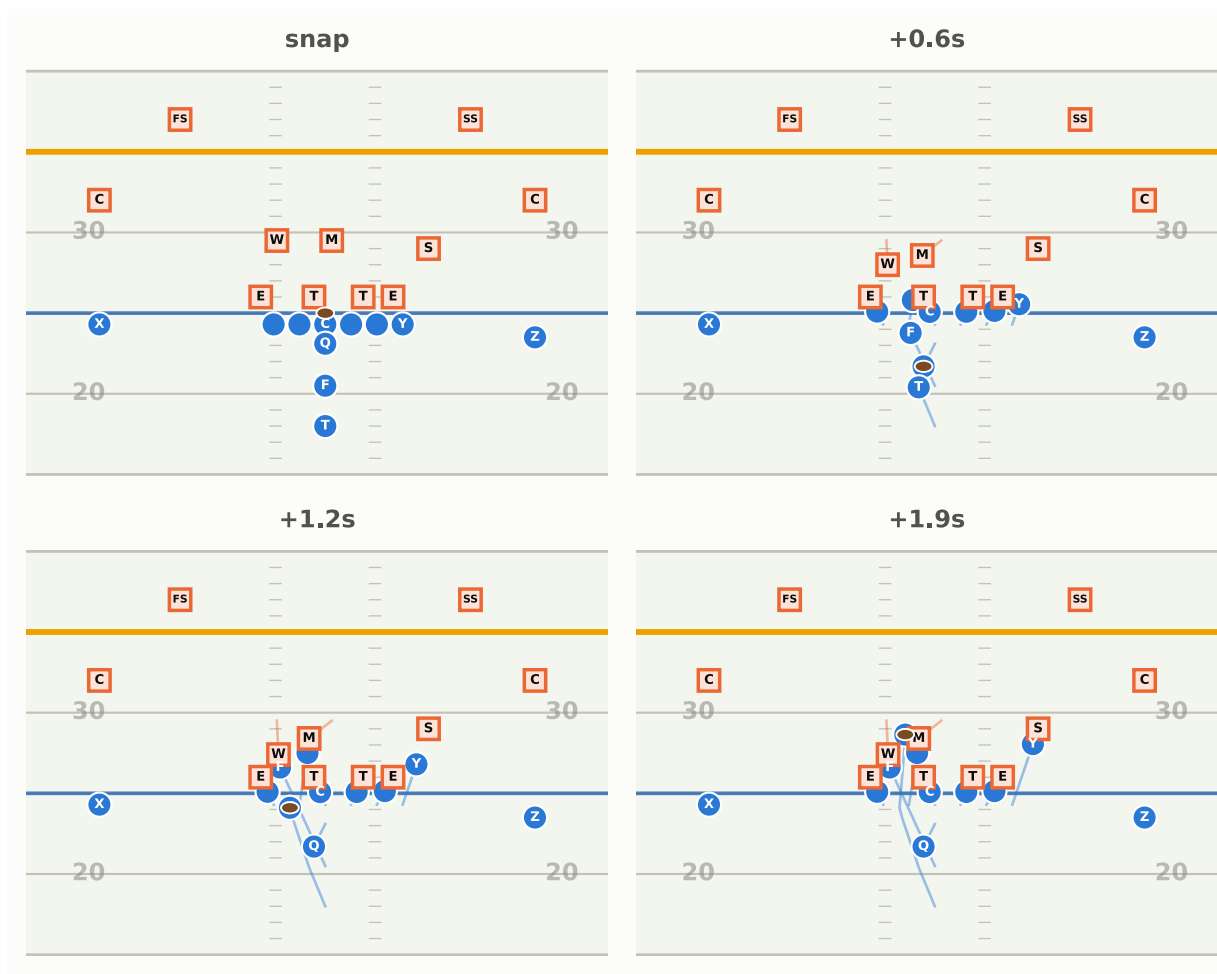


Figure 6: The iso in motion (simulated, BDB-format tracking). At the snap the fullback (F) is already the closest back to the line. By 0.6 seconds he is crossing the line while the tailback (T) takes the handoff about five yards deep. By 1.2 seconds the fullback has met the Will linebacker (W) at the line and the tailback, now at full speed, is a step behind him. At 1.9 seconds the tailback is through the hole, past the linebackers. The faint trails show where everyone has been.

What the I gives up

Every strength of the I has a price.

- **It's slow to the edge.** A tailback seven yards deep has a long way to go to get around the end. Outside runs from the I take time, and a fast defense can beat them to the sideline.
- **The backs are poor pass threats.** The fullback is in the middle of the formation and the tailback is deep; neither can get into a pass route quickly. On a pass, the I has really only three quick receivers (the two wideouts and the tight end), and the defense knows it.
- **It's obvious.** Two backs stacked behind an under-center quarterback says “run” loudly. In neutral situations from 2022 to 2025, NFL offenses lined up under center with two backs threw on only about 27% of plays.⁷ That's the price, and also, as you'll see, the opportunity.

Offset I: cheating the fullback

A fullback directly behind the quarterback has to choose a side after the snap, and the defense can't tell which side. That's good for disguise but bad for geometry: if the play goes wide, he has to run sideways before he can block anyone. The **offset I** moves the fullback a yard or two to one side, behind a guard or tackle, so he starts closer to where he's needed.

⁷Course calculation from nflverse play-by-play joined to FTN charting, regular-season runs and passes in 2022–2025, first and second down, win probability 20–80%, outside the last two minutes of each half: under center with two or more players in the backfield besides the quarterback, 27% passes (8,579 plays), of which 85% used play-action. Shotgun with one or no back: 70% passes.

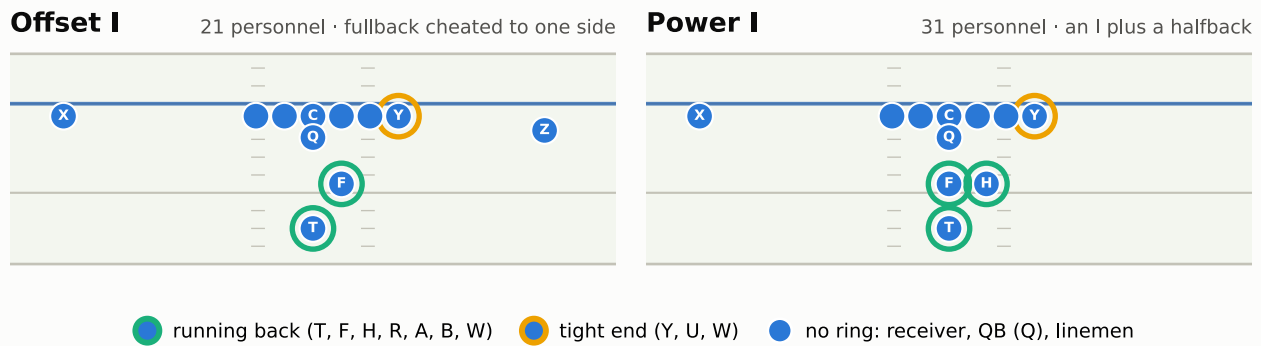


Figure 7: Two variations on the I. Left: the offset I, here offset to the strong (tight end) side, with the fullback (F) behind the right guard instead of directly behind the quarterback. Right: the Power I, which adds a third back, a halfback (H), beside the fullback, so the tailback (T) has two lead blockers in front of him. Both are under center; the Power I gives up a receiver to do it (31 personnel).

Coaches say the fullback is offset **strong** (toward the tight end) or **weak** (away from it). Offset strong, he's in position to lead on runs to the tight end side, where the offense has an extra blocker. Offset weak, he's in position to lead on runs away from the tight end, or to block a blitzing linebacker coming from the weak side on a pass. The cost is a clue. A fullback offset to one side tilts the offense toward that side, and defensive coaches study which way each opponent offsets on which plays. Smart offenses run plays both ways from both offsets so the tilt means nothing.

Power I: two lead blockers

The **Power I** adds a third back beside the fullback, so the tailback has two blockers leading him: usually 31 personnel (three backs, one tight end, one receiver), another full house. It's a short-yardage and goal-line formation, built for the moment when the offense needs one yard and the defense knows it. The extra back lets the offense outnumber the defense at the point of attack: the line blocks the defensive linemen, and the two lead backs take the two linebackers who would otherwise make the tackle. The price is that the defense, seeing three backs and one receiver, can put everyone near the line too.

Singleback and ace: trading the fullback for a tight end

Through the 1960s and '70s, defenses got better at stopping two-back formations, and in 1978 the NFL changed its rules to help the passing game (pass blockers could extend their arms, and defenders could contact receivers only within five yards of the line;⁸ 1978 and the Passing Revolution has the story). Offenses wanted more receivers on the field. The simplest place to find one was the backfield.

A **singleback** formation keeps the quarterback under center but uses only one running back, directly behind him, usually about seven yards deep. The second back is gone: his spot on the field goes to a third receiver or a second tight end. With three receivers it's 11 personnel, the NFL's most common grouping; with two tight ends it's 12.

The two-tight-end version has its own name. The **ace formation** is a singleback formation with a tight end on each side of the line and two wide receivers (12 personnel). (Some teams say "ace" for any one-back look; the broadcast usage is the two-tight-end one.)

⁸"1978 NFL season," Wikipedia: https://en.wikipedia.org/wiki/1978_NFL_season (defenders may contact receivers only within five yards of the line of scrimmage, the "Mel Blount Rule"; pass-blocking rules extended to permit extended arms and open hands).

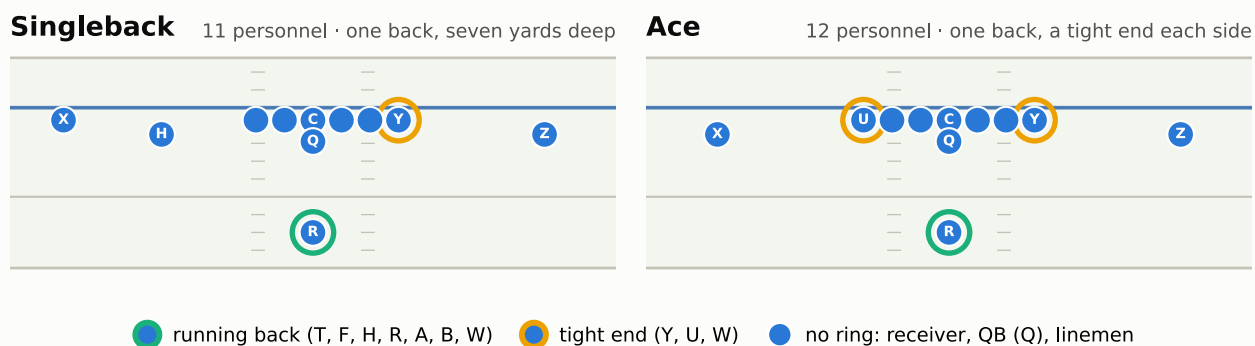


Figure 8: The one-back family. Left: singleback with 11 personnel, a tight end (Y) on the right and three receivers (X on the line, H in the slot, Z off the line). Right: the ace, 12 personnel, a tight end on each side (Y and U) and both receivers a yard off the line, so the formation is the same on both sides. In both, the lone running back (R) is about seven yards deep, directly behind the quarterback, which lets him start toward either side.

Why it exists: a tight end does a fullback's job, and more

Taking the fullback off the field sounds like giving up the lead blocker. The insight of the one-back coaches was that a **tight end can do most of a fullback's job from a different spot**, and a lot more besides. A tight end next to the tackle can block the same linebacker a fullback would have led on, or he can run a pass route down the middle, which a fullback almost never does. A second tight end on the other side makes the formation *balanced*: the same on both sides. A defense usually sets its strong side to the tight end (its biggest linebacker and a safety go there), but with a tight end on each side there isn't one, and the offense can run either way with an extra blocker.

The lone back benefits too. Seven yards deep and directly behind the quarterback, he can start toward either side and see the whole line. That's exactly what the zone running schemes of the 1990s onward wanted, a back who reads the blocks and picks his hole, and it's why the one-back, two-tight-end look became the home of the modern NFL run game (Zone Running draws it).

What did it give up? The fullback's head start. A tight end blocking from the line meets the linebacker sooner, but with less of a running start, and the tailback no longer has a blocker leading him into the hole. One-back offenses answered with motion: move a tight end or an off-the-line "H-back" across the formation before the snap so that he ends up where the lead blocker is needed.

i Film room: Washington, 1981–1992

Joe Gibbs, a Don Coryell assistant who became Washington's head coach in 1981, built the most influential one-back offense in NFL history, and won three Super Bowls with it, after the 1982, 1987 and 1991 seasons, with three different starting quarterbacks (Joe Theismann, Doug Williams and Mark Rypien).⁹ The story Gibbs told was about one player: the Giants' Lawrence Taylor, a linebacker so fast off the edge that a running back couldn't block him. Gibbs's answer was to take the second back out of the backfield and put a second tight end, often a smaller, quicker one who lined up off the line or went in motion (the *H-back*, a version of the H, or move player), next to the tackle, where he could get his hands on Taylor before Taylor got going.

The H-back became the one-back offense's Swiss army knife. Lined up off the line beside a tackle, he could block like a fullback, release like a tight end, or move across the formation before the snap and change which side had the extra blocker. Behind him, Washington's huge offensive line (the "Hogs") ran the plays that made the era famous, including the counter Trey, in which two linemen pulled the opposite way from the backfield's first step (Gap and Power Runs draws it).

What to notice: the H-back is a fullback and a tight end in one body, and he forces the defense to decide which he is on every snap.

Watch how one H-back changes the defense's picture before the snap. The offense starts with both tight ends on the right, so the defense sets its strength there. Then the H-back moves:

⁹Pro Football Hall of Fame, "Joe Gibbs": <https://www.profootballhof.com/players/joe-gibbs/> (Washington head coach 1981–1992 and 2004–2007; wins in Super Bowls XVII, XXII and XXVI); "Joe Gibbs," Wikipedia: https://en.wikipedia.org/wiki/Joe_Gibbs (Coryell's offensive coordinator in San Diego, 1979–1980). The H-back and the one-back, two-tight-end offense as Gibbs's answer to Lawrence Taylor: "H-back," Wikipedia: <https://en.wikipedia.org/wiki/H-back> (Clint Didier was Gibbs's H-back), and Tim Layden, *Blood, Sweat and Chalk* (2010).

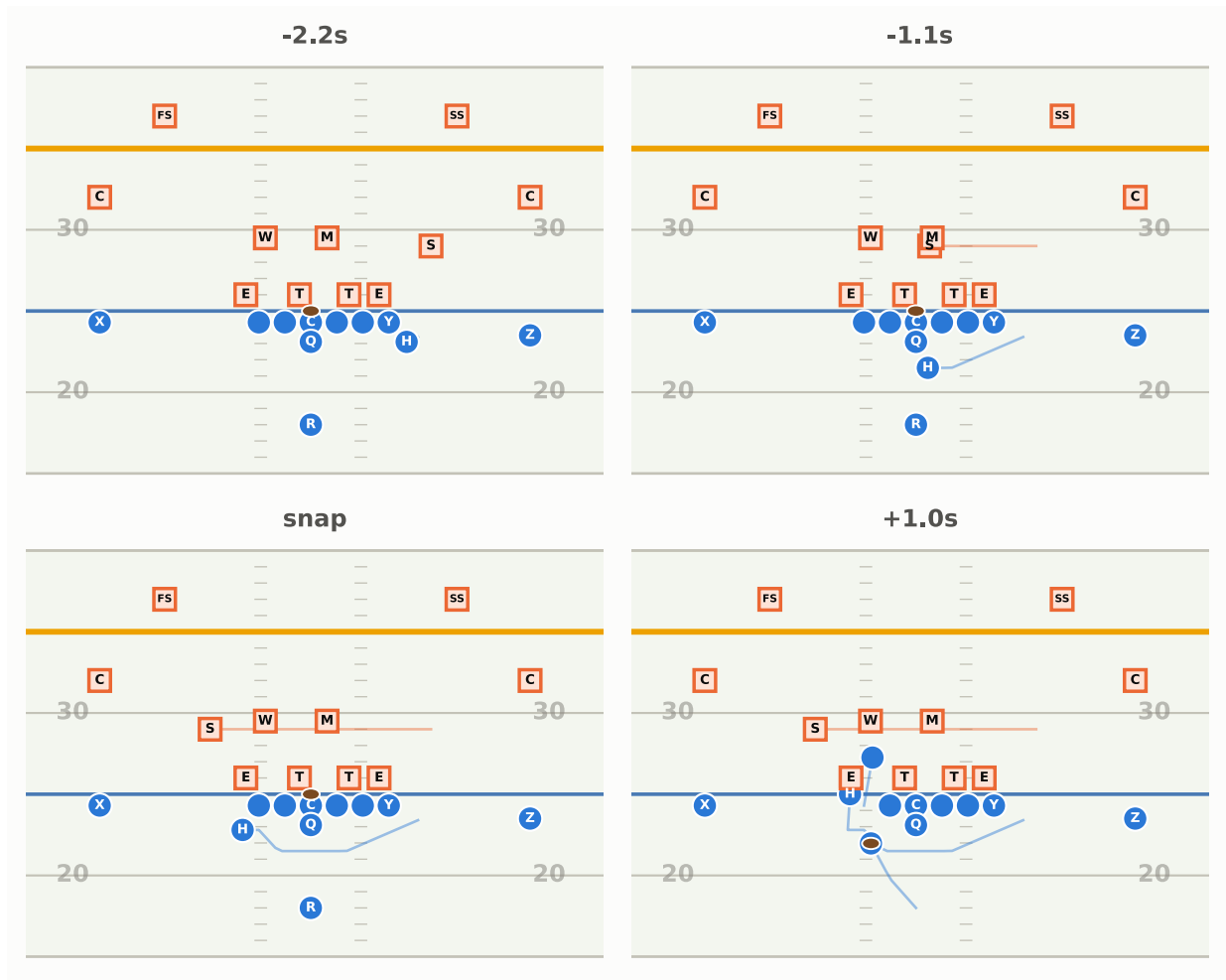


Figure 9: One-back motion (simulated, BDB-format tracking). 12 personnel: the tight end Y is on the line on the right and the H-back (H) stands just off the line outside him, so the defense's Sam linebacker (S) lines up on that side. Before the snap H moves across behind the quarterback to the left side, and the Sam has to run across with him. At the snap the offense runs to the left, with H blocking the end (E) on that side and the running back (R) following him. One player moving changed which side was strong, and which side had the extra blocker.

The option branch: wishbone and flexbone

The other way to use the T's three backs is to keep all three and let the quarterback **choose who gets the ball after the snap**. That's option football: the offense leaves a defender unblocked on purpose, the quarterback reads what he does, and gives the ball to whichever back the defender didn't take. Option football has its own chapter. Here we only need the two formations it made famous.

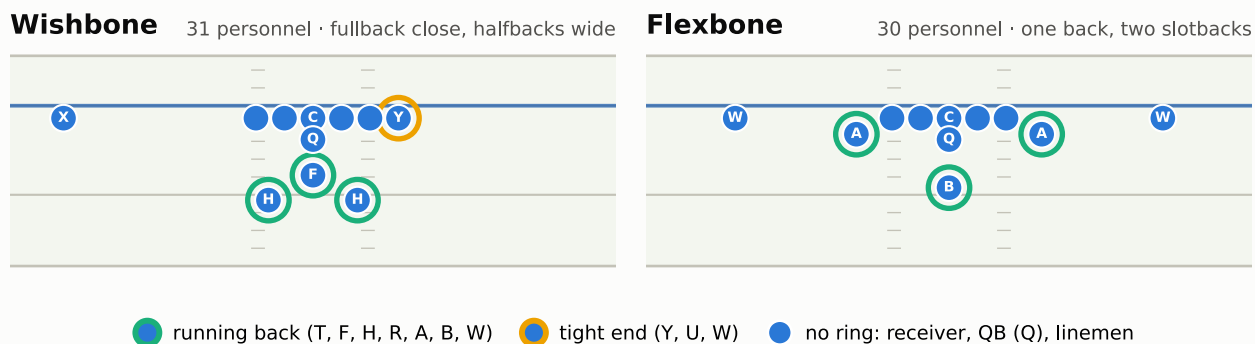


Figure 10: The option formations. Left: the wishbone (31 personnel), a full house: the fullback (F) close behind the quarterback, and two halfbacks (H) deeper and wider, one on each side, so the backs form an upside-down Y. Right: the flexbone (usually 30 personnel): one back (B) behind the quarterback, two slotbacks (A) standing a yard and a half off the line just outside the tackles, and two wide receivers (W) on the line, which makes the seven on the line legal without a tight end. The letters are the flexbone's own: A-backs and the B-back.

The wishbone

The **wishbone** was drawn up at the University of Texas in 1968 by offensive coordinator Emory Bellard, who moved the T's fullback up close behind the quarterback and set the two halfbacks deeper and wider, one on each side. A Houston sportswriter, Mickey Herskowitz, said it looked like a "pulley bone" (an old Southern name for a chicken's wishbone), and head coach Darrell Royal turned that into "wishbone."¹⁰ Texas won 30 straight games with it, Oklahoma and Alabama adopted it, and through the 1970s the wishbone was the most feared offense in college football.

The geometry is built for the triple option, the play that gives the formation its purpose. The fullback, close to the line, is the first threat (the "dive"), straight ahead. The quarterback is the second, running along the line. The halfback on the play side is the third, trailing the quarterback as a pitch man. The defense has to account for all three, on whichever side the play goes, and the quarterback decides who gets the ball by reading two defenders after the snap. The halfback on the other side becomes a lead blocker. Every back has a job on every play, on both sides; that is what a full house buys an option team. (Option Football draws the reads.)

The flexbone

The wishbone's weakness was width. With all three backs bunched behind the quarterback and only one receiver, defenses could crowd the line and outnumber the option. The **flexbone**, which grew out of the wishbone in the late 1970s and 1980s, takes the two halfbacks out of the deep backfield and stands them just off the line, outside the tackles, as *slotbacks*. A fullback stays behind the quarterback, and there are two wide receivers on the line, usually with no tight end. Who invented it is disputed: Emory Bellard himself moved a halfback up to a wing at Mississippi State in 1979 (his "wingbone"); Air Force's option teams of the 1980s, whose coach Fisher DeBerry embraced the name "flexbone"; and Paul Johnson, offensive coordinator at Georgia Southern in the mid-1980s, all have a claim.¹¹

Why there? A slotback just off the line is three things at once. He can *block* a defender on the edge from a short distance. He can *go in motion* before the snap and arrive at the pitch man's spot just as the ball is snapped, so the defense doesn't know which side the option is going. And he can *run a pass route* immediately, because he's already past the line's congestion. The two receivers outside force the defense to keep two defenders wide, which takes them out of the box. The flexbone spreads the defense like a passing formation while keeping three runners near the ball.

¹⁰"Wishbone formation," Wikipedia: https://en.wikipedia.org/wiki/Wishbone_formation (Emory Bellard, Texas offensive coordinator under Darrell Royal, 1968; Houston sportswriter Mickey Herskowitz said it looked like a "pulley bone," and Royal changed the name to "wishbone"; Texas then won 30 straight games and two national championships; Royal showed it to Bear Bryant in 1971); Texas Longhorns lettermen's history, "1968–1970: 30 straight wins and two national championships": <https://www.texaslsn.org/football-dkr-1968-1970>.

¹¹"Option offense," Wikipedia: https://en.wikipedia.org/wiki/Option_offense (the flexbone "invented by Emory Bellard at Mississippi State in 1979"; also "origins attributed to Paul Johnson while at Georgia Southern in the mid-80s"; Fisher DeBerry of Air Force "welcomed the name flexbone," while Johnson rejected it); "Wishbone formation," Wikipedia: https://en.wikipedia.org/wiki/Wishbone_formation (Bellard's "wing-bone" at Mississippi State); "Paul Johnson," Wikipedia (Georgia Southern offensive coordinator under Erk Russell, I-AA champions 1985 and 1986).

i Film room: Navy and Georgia Tech, Paul Johnson's flexbone

No coach is more closely tied to the flexbone than Paul Johnson. He won two Division I-AA national championships at Georgia Southern (1999 and 2000) with it, then took over a Navy team that had gone 0–10 in 2001 and, within two seasons, had it winning eight games and beating Army, which it did six years running under him (2002–2007). At Georgia Tech from 2008 to 2018 he ran the same offense against Atlantic Coast Conference defenses, winning the Orange Bowl after the 2014 season.¹² Navy kept the flexbone under Johnson's successor, Ken Niumatalolo (2007–2022), whose quarterback Keenan Reynolds set an FBS career record for rushing touchdowns in 2015.

Why would an undersized team choose an old formation? Because it changes the problem. The service academies can't recruit 310-pound linemen who will leave for the NFL; they can recruit disciplined, smart players. The flexbone needs linemen who can block quickly at defenders' legs and backs who carry out fakes on every play, and it makes the defense play assignment football for sixty minutes, something defenses that face it once a year rarely practice. Even Navy has since moved much of its option game into the shotgun: under coach Brian Newberry it mixed in shotgun snaps from 2023, and from 2024 offensive coordinator Drew Cronic's "Millennial Wing-T" made the shotgun a regular home for the option.¹³

What to notice: when you see a flexbone team on a Saturday, watch the slotbacks before the snap. The one who goes in motion is usually telling you which way the option is headed, and the other one is often a blocker.

You won't see either formation in the NFL. Option football needs a quarterback who takes a hit on most plays, and NFL teams won't spend their most expensive player that way; NFL defenders are also fast enough to cover the width the option relies on. But the *idea*, a quarterback who reads a defender instead of blocking him, came back to the NFL from the shotgun, as the zone read. The College Laboratory traces that lineage, from the wishbone and veer to the spread option.

The Wing-T: misdirection

The last classic branch crossed the T with its ancestor. Around 1950 Dave Nelson, first at Maine (1949–1950) and then at the University of Delaware (from 1951), combined the quick-hitting T backfield with the single wing's *wingback*, a back lined up just outside and a yard behind the tight end.¹⁴ The **Wing-T** puts the quarterback under center, a fullback directly behind him, a halfback beside the fullback on the side away from the tight end, and the wingback outside the tight end on the other side, with a split end on the halfback's side. Counted as backs, the wingback and halfback make it 31 personnel; teams that use a receiver at wingback call it 21.

¹²"Paul Johnson (American football coach, born 1957)," Wikipedia: [https://en.wikipedia.org/wiki/Paul_Johnson_\(American_football_coach,_born_1957\)](https://en.wikipedia.org/wiki/Paul_Johnson_(American_football_coach,_born_1957)) (Georgia Southern head coach 1997–2001, I-AA titles 1999 and 2000; Navy 2002–2007, 6–0 against Army; Georgia Tech 2008–2018, Orange Bowl win over Mississippi State, 49–34, Dec. 31, 2014); "2001 Navy Midshipmen football team," Wikipedia: https://en.wikipedia.org/wiki/2001_Navy_Midshipmen_football_team (0–10); "2003 Navy Midshipmen football team," Wikipedia: https://en.wikipedia.org/wiki/2003_Navy_Midshipmen_football_team (8–5, Houston Bowl); "Ken Niumatalolo," Wikipedia: https://en.wikipedia.org/wiki/Ken_Niumatalolo (Navy head coach 2007–2022, from the 2007 Poinsettia Bowl); NCAA.com, Dec. 28, 2015: <https://www.ncaa.com/news/football/article/2015-12-28/college-football-navys-reynolds-breaks-fbs-record-88th-td> (Keenan Reynolds's 88th career rushing touchdown, an FBS record).

¹³"Refined Navy offense emerging under Newberry," *Stars and Stripes*, Aug. 16, 2023: <https://www.stripes.com/sports/2023-08-16/refined-navy-offense-emerging-newberry-10996804.html> (under-center and shotgun formations, the triple option still the base); "Yes, Navy has an undefeated record, and one of the nation's most explosive offenses," Fox Baltimore, Oct. 24, 2025: <https://foxbaltimore.com/sports/sports-unlimited/navy-football-undefeated-explosive-offense-wing-t> (Drew Cronic hired after the 2023 season; the "millennial Wing-T," with the option, RPOs and shotgun snaps).

¹⁴"David M. Nelson," Wikipedia: https://en.wikipedia.org/wiki/David_M._Nelson (head coach at Maine 1949–1950, where he began developing the Wing-T, and Delaware 1951–1965); "Tubby Raymond," Wikipedia: https://en.wikipedia.org/wiki/Tubby_Raymond (Delaware head coach 1966–2001); Harold R. Raymond, *The Delaware Wing-T: An Order of Football* (Prentice Hall, 1986); Raymond's longtime assistant Ted Kempinski co-wrote later Delaware Wing-T books.

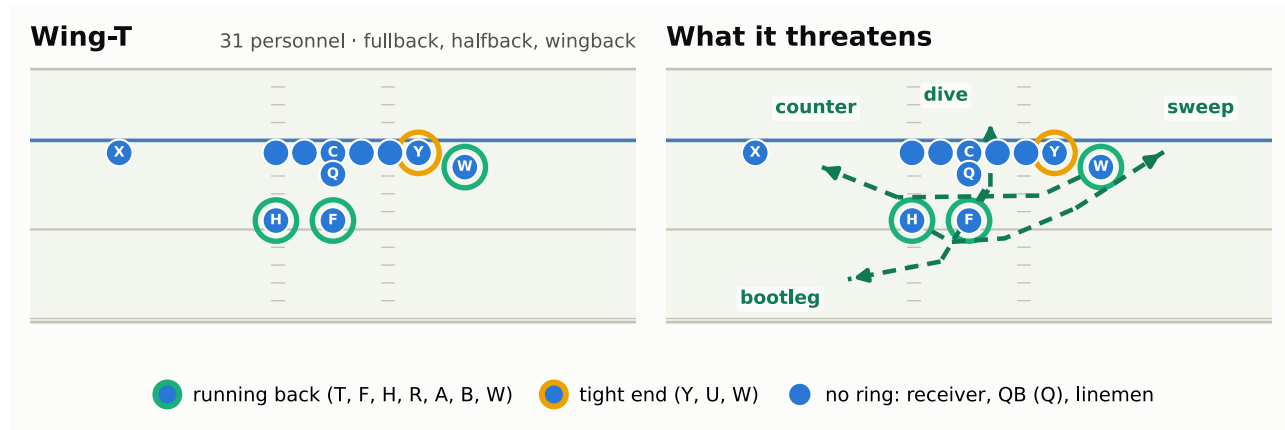


Figure 11: The Wing-T. Left: the plate: fullback (F) directly behind the quarterback, halfback (H) behind the left tackle, wingback (W) just outside and behind the tight end (Y) on the right, split end (X) on the left. Right: what the backfield threatens in the first second. The fullback can take the ball straight ahead, the halfback can run a sweep across to the wing side, the wingback can come back the other way on a reverse or counter, and the quarterback can fake to all of them and roll out away from the flow (the bootleg). Every play starts with the same motion, so the defense can't tell which back has the ball until it's too late.

Why it exists: making the defense guess

The Wing-T's purpose is misdirection: making the defense commit to one direction and then going the other. With three backs placed so that each can carry the ball in a different direction, and a quarterback who turns his back to the defense and fakes to each of them in turn, every play looks the same for its first second. The Wing-T is organized in series: a group of plays that start with the same backfield action and end differently. The famous one is the buck sweep series, in which the fullback plunges up the middle, the halfback sweeps around the wing side behind two pulling guards, and, when the defense starts chasing the sweep, the same action becomes a trap up the middle or a quarterback bootleg the other way. Defenders who flow with the first thing they see get punished; defenders who wait get blocked.

The trade is complexity and, again, width. A misdirection offense needs a lot of practice time to run its fakes well, and its backs are runners and blockers, not receivers. You'll rarely see a true Wing-T on Sunday, but you'll see its children every week: the jet sweep, in which a receiver sprints across behind the line just as the ball is snapped, and the fakes that come with it are Wing-T ideas in an NFL uniform (Perimeter Runs and Misdirection traces them).

The goal-line formation: everyone in close

Near the goal line the field gets short and narrow at the same time. There's no room behind the defense to throw deep, and the offense needs a yard or two, not twenty. So every team keeps a **goal-line formation**: the quarterback under center, two backs stacked behind him, three tight ends (or a sixth lineman, the jumbo grouping), and no wide receivers at all. Everyone lines up tight to the ball. In personnel terms that's 23, or 13 with a fullback swapped for a tight end, or a jumbo grouping.

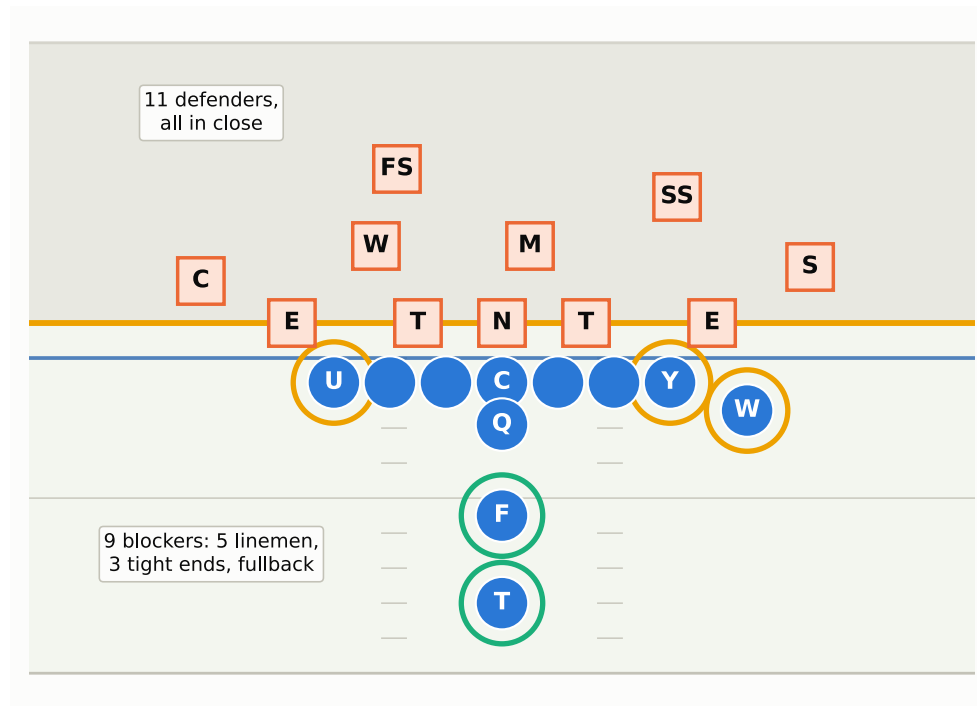


Figure 12: Second-and-goal at the 1: the goal-line formation (23 personnel) against a goal-line defense (five linemen, three linebackers, three defensive backs). Offense: three tight ends (Y and U on the line, W a wing just behind and outside Y), a fullback (F) and a tailback (T), all within about seven yards of the ball. Defense: E = end, T = tackle, N = nose tackle, S/M/W = linebackers, C = cornerback, SS and FS = safeties. Nobody on either side is more than about ten yards from the ball.

Why it exists: more blockers than defenders have gaps

The goal-line formation is arithmetic. Put eight blockers at the line (five linemen and three tight ends) and a fullback behind them, and the defense has to find a defender for every gap between and outside those blockers, plus one for the ball carrier. Tight splits (the gaps between linemen shrink to a foot or two) mean there are fewer gaps and less room to slip through them. The fullback adds one more blocker who can go wherever the defense is short. The defense answers by bringing its own heavy package and crowding the line, so the goal line is the one place in football where both sides deliberately give up all their space.

The price is that a defense that knows you'll run can commit everyone to stopping it. That's why the goal-line formation is also where a lot of short play-action touchdowns come from: the quarterback fakes the handoff, every defender bites, and one of the three tight ends, who looked like a blocker, slips into the end zone alone. In 2022–2025, about three of every four NFL plays inside the 2-yard line that came from under center with two or more backs were runs; when they were passes, about four in five used a play fake.¹⁵ Red Zone, Goal Line, and Short Yardage is where that fight gets its full treatment.

Matching formations to personnel

Personnel comes first, so the formation is usually a short list once you've counted the backs and tight ends. The table below is the cheat sheet. The pairings are typical, not laws: a team can line up 21 personnel in the shotgun, or 12 personnel in an I with a tight end at fullback.

Formation	Typical personnel	Backfield	Built for	Where you'll see it now
T-formation	32 (or 22)	Three backs side by side	Quick hitting runs to either side	Goal line (as a full house), rarely
Pro set (split backs)	21	Two backs side by side	Balance: run either way, backs in the flats	Occasionally, mostly play-action

¹⁵Course calculation from nflverse play-by-play joined to FTN charting, regular season 2022–2025, plays with the ball at the opponent's 1 or 2: under center with two or more players in the backfield besides the quarterback, 23% passes (825 plays), and 79% of those passes used play-action (73% of all under-center passes inside the 2).

Formation	Typical personnel	Backfield	Built for	Where you'll see it now
I-formation	21 (or 22)	Two backs stacked	Lead blocker, downhill tailback	21-personnel teams; short yardage
Offset I	21 (or 22)	Fullback a yard or two to one side	Better angles for the lead block	Same as the I
Power I	31	I plus a halfback	Two lead blockers	Goal line and short yardage
Singleback	11	One back at ~7 yards	Three receivers with an under-center run game	Every NFL team, every week
Ace	12	One back at ~7 yards	Balance, a tight end blocker on both sides	Every NFL team, every week
Wishbone	31 (or 32)	Fullback close, halfbacks deep and wide	The triple option	History, some high schools
Flexbone	30	One back, two slot-backs	Option with width	Service academies, high schools
Wing-T	31 (or 21)	Fullback, halfback, wingback	Misdirection, series football	High school; its ideas in the NFL
Goal-line	23 (or 13, or jumbo)	Two backs stacked	Raw blocking numbers	Every NFL team, inside the 3

And the reverse direction, from what you count to what you should expect:

- **One back, quarterback under center:** singleback (11) or ace (12).
- **Two backs, quarterback under center:** I, offset I or pro set (21 or 22). Look at whether they're stacked, offset or side by side.
- **Three backs:** Power I, wishbone or T (31 or 32), almost always short yardage in the NFL.
- **No wide receivers and three tight ends:** goal-line.
- **Quarterback in the shotgun:** you're in the next chapter's world, even with two backs.

Common misconception

"Formation names are universal." They're universal enough for broadcasts, but not in playbooks. One team's "ace" is another's "Tiger" or "Heavy"; one team calls split backs "Red" and "Blue," another calls them "Pro" and "Split." Even the broadcast names blur: an offset I with a tight end at fullback, or a singleback with the back in the pistol, may be called three different things by three analysts. Learn the shapes, not just the words: under center or not, how many backs, where they stand, where the tight ends are. The shape is what tells the defense, and you, what's possible.

Why the classic formations faded

For most of the twentieth century, the quarterback took the snap under center on nearly every play. Today he does it on about a third of them. Here's what that decline looks like in the data.

648,482 regular-season runs and passes 2006-2025; 133,636 of them 2022-2025

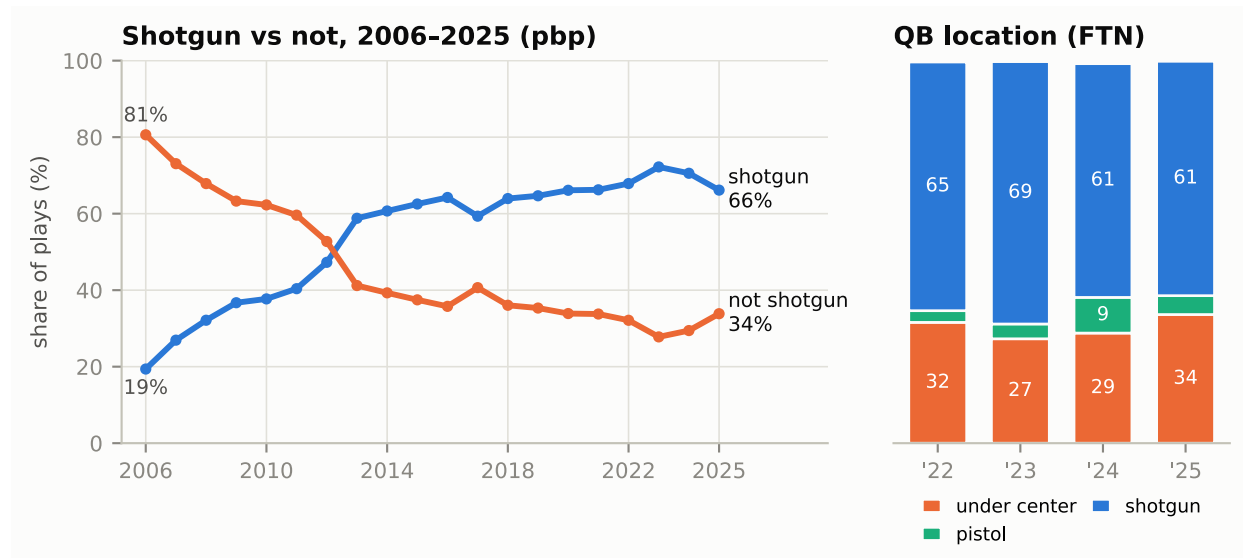


Figure 13: The quarterback moves back. Left: share of NFL regular-season runs and passes snapped to a quarterback in the shotgun, 2006–2025 (nflverse play-by-play `shotgun` flag; the rest are under center or, in recent seasons, a few percent from the pistol). Right: where the quarterback lined up, 2022–2025, from FTN’s play-by-play charting (which records under center, shotgun and pistol separately). The shotgun peaked in 2023 and fell back in 2025, when under center rose to about a third of plays. Different sources, so different panels.

In 2006 offenses were in the shotgun on about one play in five. By 2023 it was about 72%. Over the same years, the NGS formation labels in the participation data show the I-formation falling from about 11% of plays in 2016 to about 6% in 2022.¹⁶ The causes overlap, but there are five worth knowing.

1. **Passing became more valuable than running.** After the 1978 rule changes, and again as offenses learned to throw efficiently, a pass gained more on average than a run. Expected Points and Success Rate shows how analysts measure that gap. An offense that passes more wants more receivers, and the second back is the first player to go.
2. **The shotgun is a better place to pass from.** Five yards back, the quarterback doesn’t have to turn his back on the defense or take a three- or five-step drop. He sees the coverage from the snap and has more time before the rush arrives. The shotgun was used in the NFL as early as 1960 and revived by Tom Landry’s Cowboys in 1975, but for decades it was a third-down formation;¹⁷ the history chapters follow it.
3. **Spreading the defense makes running easier, too.** Three or four receivers pull defensive backs out wide, which leaves fewer defenders near the ball. An offense in the shotgun with 11 personnel can run against six defenders in the box instead of eight. Spread formations are the next chapter.
4. **College stopped producing under-center quarterbacks.** By the 2010s most college offenses, and many high schools, played almost entirely from the shotgun. Quarterbacks arrived in the NFL having rarely taken a snap under center, and NFL coaches built around what their quarterbacks did best.
5. **A fullback is an expensive specialist.** A roster spot used on a lead blocker is one not used on a receiver or tight end who can play every down. Many NFL teams stopped carrying a true fullback at all.

¹⁶Course calculation from nflverse participation data (`offense_formation`, supplied by Next Gen Stats through 2022), regular-season runs and passes: I_FORM 10.7% of plays in 2016, 6.2% in 2022. From 2023 the source changes to FTN, which records only SHOTGUN, UNDER CENTER and PISTOL, so the series can’t be extended. The shotgun shares in the chart are from the nflverse play-by-play `shotgun` flag; the quarterback-location shares from FTN charting (`qb_location`), joined to play-by-play: https://nflreadr.nflverse.com/articles/dictionary_ftn_charting.html

¹⁷Pro Football Hall of Fame, “The Shotgun Formation”: <https://www.profootballhof.com/news/the-shotgun-formation> (Red Hickey’s 49ers first used it on Nov. 27, 1960, against the Colts; Tom Landry’s Cowboys revived it in 1975 to protect Roger Staubach); “Shotgun formation,” Wikipedia: https://en.wikipedia.org/wiki/Shotgun_formation.

! Why it exists

The shotgun won the passing game; the classic formations kept the running game's advantages. Under center, the quarterback can hand off to a back already running downhill, toward the line, and fake a handoff with his back to the defense. In the shotgun, many runs start sideways, and a play fake is less convincing because the defense can see the ball the whole time. Each formation is a bundle of trade-offs, and the NFL's long move to the shotgun traded the run game's angles for the passing game's vision. The 2025 turn back toward under center is offenses deciding they gave up too much.

Where the classic formations survive

The I-formation isn't dead. It has retreated to the places where its advantages matter most, and in 2025 it started coming back out. Look at what lining up under center tells you now:

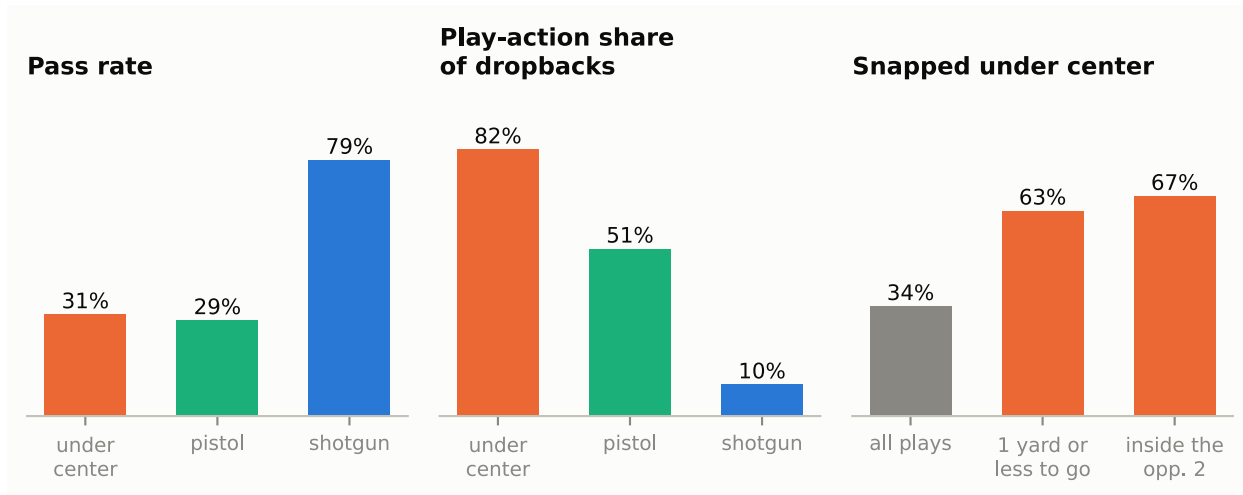


Figure 14: What lining up under center means in the 2025 NFL (regular season, FTN charting via nflverse). Left: how often the offense passed from each quarterback location. Middle: of the dropbacks from each location, the share that used a play-action fake. Right: the share of plays snapped under center, overall and on short yardage (1 yard or less to go) and inside the opponent's 2-yard line. Under center is a running alignment, it is the home of play-action, and it is the default when the offense needs a yard.

Three things stand out, and they're the three places the classic formations live.

Short yardage. With a yard or less to go, NFL offenses lined up under center on about 63% of plays in 2025, nearly twice their overall rate. When the offense needs one yard, the quick snap, the downhill handoff and the extra blockers of the I and the goal-line formation are worth more than the quarterback's view of the coverage. (The quarterback sneak, and its push-assisted cousin, start from under center too.)

Play-action. About four of every five dropbacks from under center in 2025 used a play fake, against about one in ten from the shotgun. That's no coincidence. A quarterback who turns his back to the defense and puts the ball in a running back's belly is selling the run with his whole body, and a two-back formation that usually runs is the best advertisement for a fake. The formation's predictability is the bait. Play-Action, Bootlegs, and Screens shows what the fake does to the defense.

21 personnel. A few NFL teams still build around a fullback, and they're among the most interesting offenses in the league.

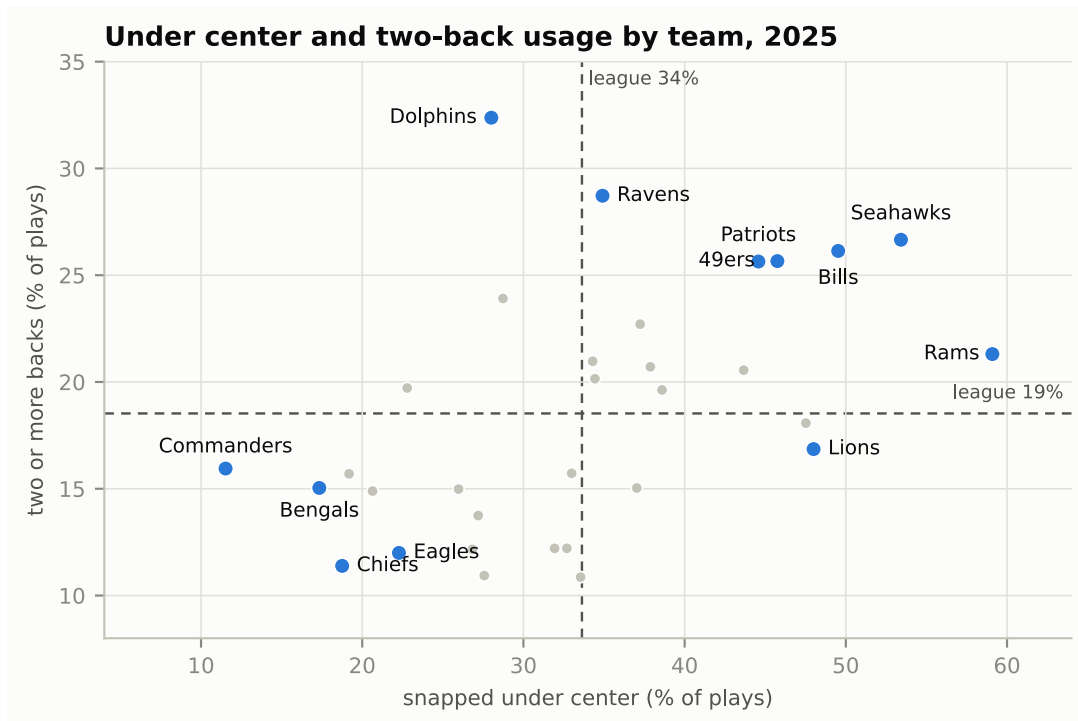


Figure 15: Who still plays classic football: each NFL team’s 2025 share of plays snapped under center (across) against its share with two or more players in the backfield besides the quarterback, usually two backs (up). Dashed lines are the league rates. The Rams, Seahawks and Bills led the under-center revival; the Dolphins, Ravens, Seahawks, Bills, Patriots and 49ers used the most two-back sets. Regular season, runs and passes; FTN charting via nflverse.

The two axes don’t always go together, which is the interesting part. The Rams were the league’s most under-center team in 2025 (about 59% of plays) but used two backs only a little more than average; their under-center snaps were mostly one-back looks with two or three tight ends. The Dolphins were the opposite: more two-back plays than anyone, mostly from the shotgun and pistol. The teams in the top right, the 49ers, Seahawks, Bills and Patriots, are the ones still playing something close to classic football, a fullback and a tailback behind a quarterback under center, on a quarter of their snaps or more.

i Film room: San Francisco 49ers, 2017–2025

Kyle Shanahan’s 49ers have been the NFL’s leading 21-personnel team for most of his tenure, with fullback Kyle Juszczyk (Personnel has the numbers). Much of it comes from the I and offset I: Juszczyk lines up behind the quarterback, leads on a linebacker on running plays, and on play-action slips out into the flat as the defense crashes on the run. In 2025 the 49ers were under center on about 46% of plays, well above the league’s 34%, and had two backs on the field on about a quarter.¹⁸

The 49ers’ version of the I isn’t the 1970s version, though. Juszczyk motions out of the backfield, lines up as a wing or in the slot, and returns to the I on the next play, so the defense can’t key on where he stands. The formation is classic; the way it’s used is modern.

What to notice: when the 49ers line up in the I, watch the fullback’s first step. If he heads for a linebacker, it’s a run; if he heads for the flat while the quarterback turns his back, the defense is about to be fooled by its own expectations.

¹⁸Course calculation from nflverse play-by-play joined to FTN charting, 2025 regular season: 49ers under center on 45.8% of runs and passes, two or more backs on 25.7%; league 33.6% and 18.5%. Juszczyk’s role and record: see The Twenty-Two.

The 2024 Ravens (head coach John Harbaugh, offensive coordinator Todd Monken, an era that ended after 2025) paired Lamar Jackson with Derrick Henry, who ran for 1,921 yards, the second-highest total in the league, behind a fullback who had started his career as a defensive lineman.¹⁹ Patrick Ricard, an undrafted defensive tackle from the University of Maine, became one of the NFL's few full-time fullbacks: a 300-pound lead blocker for a 247-pound tailback. Baltimore had two or more backs on the field on about a quarter of its 2024 plays, against a league rate of about 18%.

The Ravens show what an I-formation does to a modern defense. A defense built to stop Jackson's passing and scrambling sits in nickel; the Ravens line up with two backs and a tight end, and the nickel back suddenly has a 300-pound fullback coming at him. If the defense answers with linebackers, Jackson throws over them. That is the personnel dilemma, delivered from a formation your grandfather would recognize.

What to notice: in a classic two-back formation, find the fullback and count the defenders in the box. If the defense left a defensive back where a linebacker should be, the offense has the matchup it wanted.

The 2025 under-center revival

The shotgun's share fell from 72% in 2023 to 66% in 2025, and FTN's charting has offenses under center on 33.6% of plays in 2025, up from 27.3% in 2023.²⁰ The reason is the same arms race that sent personnel heavier (Personnel told that half): defenses spent the early 2020s getting lighter and playing more two-deep safety looks to stop the pass, which left fewer big bodies near the line. Offenses answered by getting bigger and putting the quarterback back under center, where the run game has its best angles and play-action its best disguise. Sean McVay's Rams were under center on about 59% of their 2025 snaps, the most in the league, and their quarterback, Matthew Stafford, was the league's MVP. The champion Seahawks, under offensive coordinator Klint Kubiak in 2025, were second, at about 53%.

Whether the turn lasts depends on the defense's answer, and in 2025 defenses were already answering by playing more base. The old formations didn't come back because they were old. They came back because, for now, they're the right answer to what defenses are doing. That's the pattern of this entire book.

The classic formations on one page

Here are all eleven, reduced to what you'll actually see on a broadcast: the line, the quarterback and the backfield. The backfield is the fingerprint, so it's what to memorize.

¹⁹Pro Football Reference, Derrick Henry: <https://www.pro-football-reference.com/players/H/HenrDe00.htm> (1,921 rushing yards in 2024, second to Saquon Barkley's 2,005; listed at 247 lb); "Patrick Ricard (American football)," Wikipedia: [https://en.wikipedia.org/wiki/Patrick_Ricard_\(American_football\)](https://en.wikipedia.org/wiki/Patrick_Ricard_(American_football)) (listed at 300 lb; undrafted from Maine in 2017 as a defensive lineman; with Baltimore 2017–2025, then the Giants). Two-back shares (Ravens 24.3%, league 17.7%, 2024 regular season, two or more players in the backfield besides the quarterback) are a course calculation from FTN charting via nflverse. The 2024 staff and its end after 2025 are from the course fact sheet (Harbaugh fired January 2026).

²⁰Shotgun share from nflverse play-by-play (72.2% in 2023, 66.2% in 2025) and under-center share from FTN charting (27.3% in 2023, 33.6% in 2025), regular-season runs and passes, as checked in the course fact sheet (FACTS-current §10); the team figures are the course's own calculation from the same data (Rams 59.1%, Seahawks 53.4% under center in 2025). Stafford's MVP (AP, Feb. 5, 2026) and Seattle's 2025 offensive coordinator are from the same fact sheet.

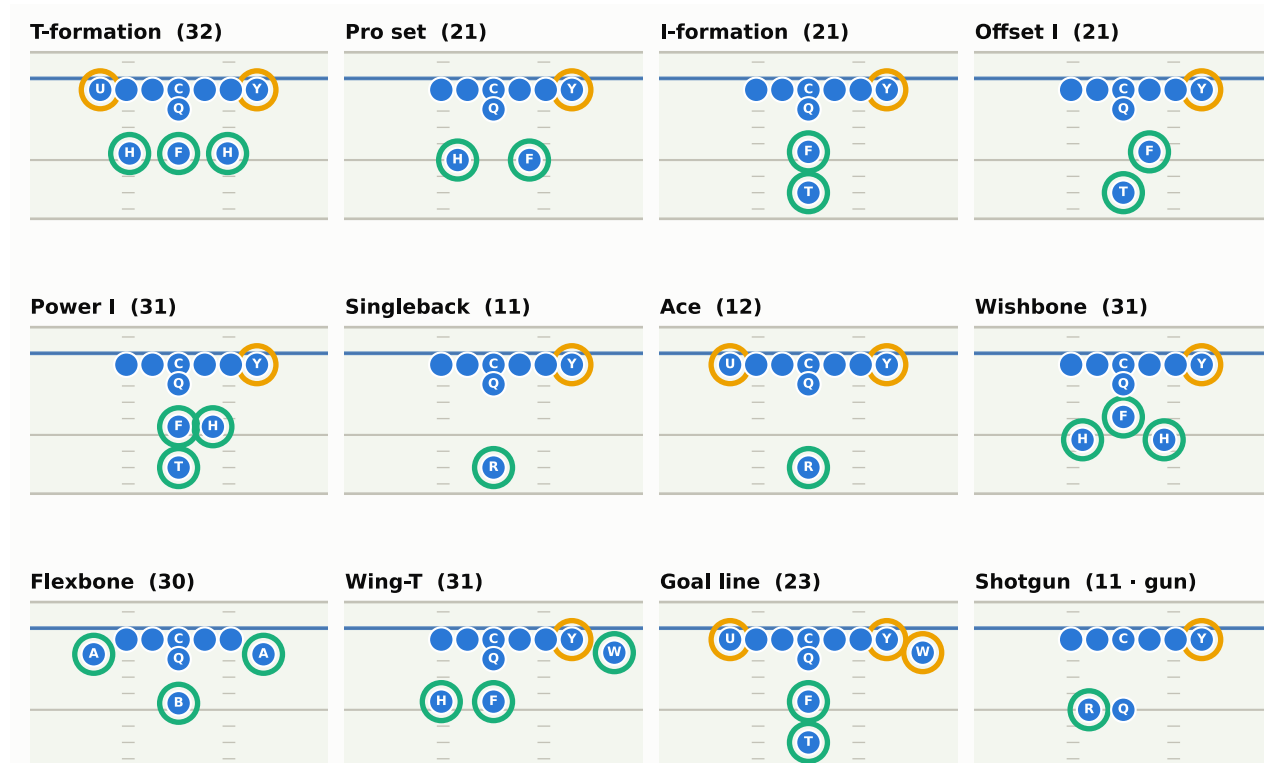


Figure 16: The eleven classic backfields, with the shotgun for comparison (last panel). Only the players within about eight yards of the ball are shown: the line, the quarterback (Q), the backs (aqua rings) and any tight ends or wings (gold rings). Read them back to front: under center or not, how many backs, stacked, side by side or offset.

Watch for it on Sunday

💡 Watch for it: name that formation

For ten snaps in a row, before the ball is snapped, say three things out loud and then a name.

- **Under center or gun?** Look at the quarterback's hands. Under center means this chapter's formations; pistol or shotgun means the next chapter's.
- **One back or two (or three)?** Count the players behind the quarterback. A tight end standing off the line next to a tackle (an H-back) counts as a blocker, not a back, for naming.
- **Stacked, split or offset?** Two backs in a line is an I; side by side is a pro set; one a yard to the side is an offset I. Three backs is a Power I, a wishbone or a T, and in the NFL that means short yardage.
- **Where are the tight ends?** One on each side with one back is an ace. Three tight ends and no receivers is goal line.
- **Name it, then guess.** Under center with two backs: lean run, and if it's a pass, expect a play fake. Under center with one back: closer to even, still play-action heavy. Note which way an offset fullback points and whether the play went that way.
- **Log it.** Add a formation column to your prediction log next to personnel: "21, I, UC" or "12, ace, UC." After ten snaps, check how often your run/pass guess was right from under center versus from the gun.

Predict the play

💡 Predict the play 1: the offset fullback

Second-and-4 at your own 40, first quarter, tied. The offense has come out like this against a base 4-3:

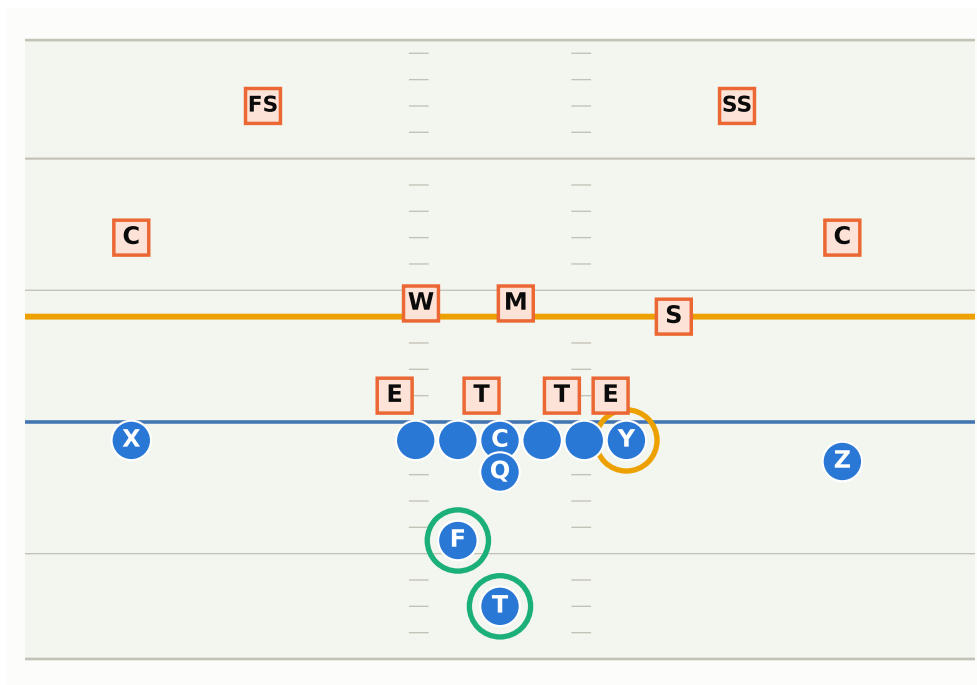


Figure 17: Drill 1: name the formation and personnel. The tight end (Y, gold ring) is on the right; the fullback (F) stands behind the left guard, the tailback (T) seven yards deep behind the quarterback. Defense: E = end, T = tackle, S/M/W = linebackers, C = cornerbacks, FS/SS = safeties.

What formation and personnel is this? Run or pass, and if it's a run, which way does the formation lean?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: second-and-goal at the 1

The offense has the ball on the defense's 1-yard line, second down. It sends in three tight ends, a fullback and a tailback; the defense answers with its goal-line package.

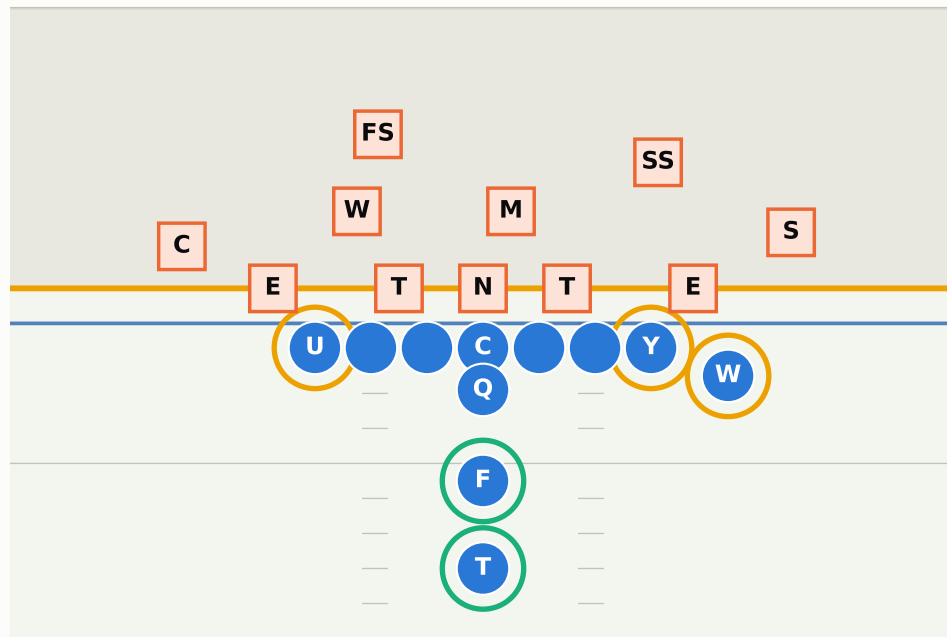


Figure 18: Drill 2: second-and-goal at the 1. The offense is in its goal-line formation; the tight end W is a wing behind and outside Y. The defense has five linemen, three linebackers and three defensive backs; its two safeties (SS, FS) are only four or five yards deep.

Run or pass? If it's a pass, who's the most likely target, and why?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: the I against eight in the box

First-and-10 at midfield, second quarter. The offense is in the I with 21 personnel. The defense, in its base 4-3, has walked the strong safety down beside the linebackers, leaving one safety deep.

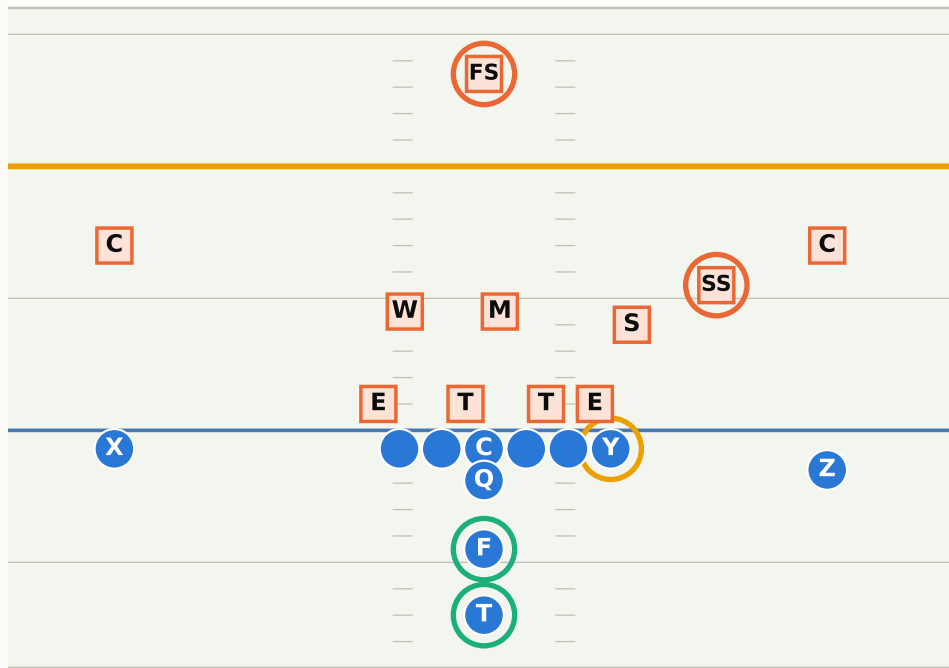


Figure 19: Drill 3: I-formation (21 personnel) against a base 4-3 with the strong safety (SS) walked down outside the Sam linebacker: eight defenders near the line of scrimmage and one deep safety (FS) in the middle of the field. The cornerbacks (C) are about seven yards off the two receivers.

Why did the defense bring the safety down, and what is the offense likely to try?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **The classic formations are defined by the backfield.** Read them back to front: quarterback under center or not, how many backs, and whether they're stacked (I), side by side (pro set), offset (offset I) or three wide (T, wishbone, Power I).
- **Each formation is an answer to a problem, and every answer has a price.** The I gives the tailback a lead blocker and a running start but is slow to the edge and obvious; the pro set is balanced but angled; the singleback and ace trade the fullback for a tight end or receiver; the wishbone and flexbone are built to read a defender instead of blocking him; the Wing-T lives on misdirection; the goal-line formation trades all its space for blockers.
- **They're one family.** The T's three backs were stacked (I), split (pro set), spread (flexbone), scattered for misdirection (Wing-T) or traded for receivers (singleback). Knowing the tree helps you name the shape.
- **Formation follows personnel.** 21 usually means an I, offset I or pro set; 12 an ace; 11 a singleback or a shotgun look; three backs or three tight ends mean short yardage.
- **Two-back, under-center football faded because passing won, and it survives where running and faking matter most.** In 2025, under center meant run about 70% of the time, play-action on about four of five dropbacks, and it was the default on short yardage. The under-center share rose to about a third of plays as offenses got bigger against lighter defenses.

Connections

Builds on: Personnel (who is on the field, and why 21 and 12 personnel matter), Formation Rules and the Language of Alignment (seven on the line, the X/Y/Z letters, under center, shotgun and pistol, strong and weak side), and The Twenty-Two (running backs, fullbacks and tight ends).

Sets up: Spread Formations (the shotgun branch of the tree), A First Look at Defense (the box and base defense, the defense's answer to these formations), and Motion and Shifts as Weapons (what the H-back's motion becomes). Later: The Language of the Run Game and Gap and Power Runs (the lead block, iso and counter trey), Option Football (the triple option from the flexbone), Perimeter Runs and Misdirection (the Wing-T's series), Play-Action, Bootlegs, and Screens, and the history in The Single Wing to the Scoring Drought, 1978 and the Passing Revolution and The College Laboratory (the wishbone and flexbone lineage).

Case studies: the Shanahan tree (the 49ers' fullback and the under-center revival) and the Ravens with Lamar Jackson.

Go deeper

- Tim Layden, *Blood, Sweat and Chalk* (2010): readable chapters on the T, the wishbone, the one-back offense and the Wing-T, built from interviews with the coaches who drew them.
- Vince Lombardi with W. C. Heinz, *Run to Daylight!* (1963): a week inside the Packers, with Lombardi explaining his sweep in his own words.
- Chris B. Brown, *The Art of Smart Football* (2015) and his Smart Football site (<https://smartfootball.com>): essays on the I, the flexbone and the shotgun's takeover.
- The nflverse FTN charting dictionary, for the quarterback-location and backfield fields used in this chapter's charts: https://nflreadr.nflverse.com/articles/dictionary_ftn_charting.html.

Answers to Predict the play

Answer 1.

An offset I, offset weak, with 21 personnel. Two backs (fullback and tailback) and one tight end, so two receivers. The fullback is offset to the left, away from the tight end: that's "weak."

Run or pass: lean run. On first and second down in neutral situations from 2022 to 2025, NFL offenses under center with two backs ran on about 73% of plays, and second-and-4 is a comfortable running down. If it's a pass, expect a play fake: about 85% of those dropbacks used one.

Which way: the offset fullback is closer to the left side, so he's in position to lead on a run to the left (a weak-side iso or lead), or to block a blitzing linebacker from that side on play-action. A defense that has studied this offense knows whether it actually runs toward its offset, and smart offenses vary it so the clue means little. Log it: whether the play went toward the fullback is exactly the kind of tendency that's worth tracking for a team over a game.

Answer 2.

Probably a run, but this is where the best play-action passes in football come from. From 2022 to 2025, NFL plays inside the 2-yard line that started under center with two or more backs were runs about three times in four. The offense has nine blockers against eleven defenders who are all within a few yards of the ball: it wants to win with numbers, and a fullback leading into a gap on a tailback dive or a quarterback sneak is the obvious call.

If it's a pass: about four in five of those passes used a play fake. The likeliest target is a tight end, often the wing (W) or the tight end on the side away from the fake: he blocks for a count, as he would on a run, then slips into the end zone while the linebackers and safeties step up to stop the run. The defense's dilemma is that every defender it keeps back to cover a tight end is one fewer to stop the run, and the offense knows it. Watch the safeties: if both crash on the snap, the tight end is alone.

Answer 3.

The defense brought the safety down because the formation says run. Two backs and a quarterback under center ran about 73% of the time in neutral situations in recent seasons. With the strong safety in the box, the defense has eight players near the line against the offense's seven blockers (five linemen, the tight end and the fullback), so it has one more defender than the offense has blockers. That's the defense's answer to the I's extra blocker.

What the offense is likely to try: the run is still the most common call, often aimed away from the extra defender. But this is the picture play-action was invented for. With eight defenders reacting to a run fake and only one safety deep in the middle, a well-sold fake can leave the outside receivers one-on-one with cornerbacks and nobody behind them on one side of the field. Under center with two backs, if the offense throws, it fakes the handoff almost every time. If you see the quarterback put the ball in the tailback's belly and pull it out, look downfield right away at the receiver on the side away from the deep safety.